

Email Module (Service Layer)

Async, firm-aware, templated email sending.
No direct API endpoints — called internally by other services.

Architecture

- **EmailService** — `@Async` service that sends emails in the background. Never blocks the main business logic.
- **FirmEmailConfig** — Per-firm SMTP settings configured by Firm Admin via `/api/v1/firm/email-config`
- **SystemConfig** — Global SMTP fallback configured by Super Admin via `/api/v1/super-admin/config`

SMTP Resolution Priority

1. **Firm's own SMTP** — If the firm has `FirmEmailConfig` with `isActive=true`, use it
2. **Global system SMTP** — Platform-level SMTP from SystemConfig (GLOBAL scoped)
3. **Spring auto-configured** — If neither is configured, Spring's default mail sender is used

Email Templates

All templates are HTML with Thymeleaf and use the firm's branding (brand color, logo, footer text, signature).

Template	Triggered By	Subject
WELCOME_EMPLOYEE	EmployeeService.createEmployee()	"Your account at [Firm Name]"
WELCOME_CLIENT	ClientService.createClient()	"Welcome to [Firm Name] Client Portal"
WELCOME_FIRM_ADMIN	FirmService.createFirm()	"Your firm [Firm Name] is ready"
PASSWORD_RESET	UserManagementService.resetPassword()	"Your password has been reset"
MFA_ENABLED	AuthService.confirmMfaSetup()	"MFA has been enabled on your account"

Configuration

Per-Firm SMTP (Firm Admin)

```
PUT /api/v1/firm/email-config
{
  "smtpHost": "smtp.gmail.com",
  "smtpPort": 587,
  "smtpUsername": "apex@apexlaw.com",
  "smtpPassword": "app-password",
  "fromName": "Apex Law Associates",
  "fromAddress": "noreply@apexlaw.com",
  "useTls": true,
  "isActive": true
}
```

Global SMTP Fallback (Super Admin)

```
PUT /api/v1/super-admin/config
{
  "SMTP_HOST": "smtp.gmail.com",
  "SMTP_PORT": "587",
  "SMTP_USERNAME": "platform@nepalcrm.com",
  "SMTP_PASSWORD": "encrypted-password",
  "SMTP_FROM_NAME": "NepalCRM Platform"
}
```

Password Security: SMTP passwords are stored AES-256 encrypted in the database. The password is never returned in API responses — only a `smtpPasswordSet` boolean flag is included. For Gmail, use App Passwords (Settings > Security > App Passwords), not the account password.